

KALABE KEBEDE

Senior Software Engineer | Forward Deployed Engineer

Maryland | (240) 288-1031 | kalabkebe12@gmail.com

GitHub: github.com/Kalab21 | Portfolio: developer-portfolio-iota-ten.vercel.app

PROFESSIONAL SUMMARY

Senior Software Engineer and Forward Deployed Engineer with 6+ years of experience building, modernizing, and supporting enterprise systems across banking, financial services, insurance, analytics, and AI-enabled applications. Strong in Java, Spring Boot, distributed systems, Apache Kafka, REST APIs, SQL, AWS, data engineering, and production delivery, with recent hands-on work in Python, FastAPI, RAG, AI evaluation, and brownfield modernization. Experienced across requirements analysis, system design, secure implementation, automated testing, observability, CI/CD, release, and production support.

TECHNICAL SKILLS

Software & Distributed Systems: Java 8-21, Spring Boot, Spring MVC, Spring Security, Spring Data JPA, REST APIs, Microservices, Apache Kafka, Event-Driven Architecture, API Gateway, Concurrency, Idempotency, Transaction Integrity

Cloud & DevOps: AWS, EC2, ECS/Fargate, S3, Lambda, RDS, Athena, Glue, CloudWatch, Docker, Kubernetes, Nginx, GitHub Actions, Jenkins, Maven, CI/CD

Data Engineering & Analytics: Advanced SQL, PostgreSQL, Oracle, MySQL, MongoDB, Redis, ETL/ELT, Data Pipelines, CDC, Data Quality, Query Optimization, Indexing, Dimensional Modeling, Power BI, Tableau, DAX, Power Query

Full-Stack, AI & FDE: Python, FastAPI, Next.js, React, TypeScript, RAG, LLM Integration, AI Evaluation, Human-in-the-Loop, Brownfield Analysis, Requirements Synthesis, Spec-Driven Development, ADRs

Security, Reliability & Testing: OAuth2/OIDC, JWT, RBAC, OWASP Concepts, Authorization Boundaries, Secrets Management, Structured Logging, Metrics, Tracing, Reconciliation, JUnit 5, Mockito, Pytest, Playwright, Integration Testing, TDD

PROFESSIONAL EXPERIENCE

Revature | Forward Deployed Engineer

May 2026 - Sep 2026

Remote

- Modernized inherited multi-service consumer-lending and skills-assessment platforms by tracing workflows across APIs, services, databases, authorization boundaries, and operational controls while delivering verified changes without breaking existing contracts.
- Built a grounded RAG policy assistant using Python and FastAPI while keeping deterministic lending decisioning as the system of record; strengthened RBAC, decision finality, idempotency, auditability, reconciliation, and data-integrity controls.
- Translated business and regulatory requirements into specifications and ADRs; implemented automated verification, structured logging, metrics, tracing, and CI/CD quality gates.
- Extended a FastAPI/Next.js skills-assessment platform with participant, trainer, scoring, reporting, analytics, and role-aware workflows backed by PostgreSQL and MongoDB, with Nginx and API-gateway routing.

Lovelytics | Data & Analytics Engineer - Contract

Dec 2024 - Nov 2025

Arlington, VA

- Designed SQL-based analytics, ETL/ELT, and reporting solutions using PostgreSQL, MySQL, Power BI, and dimensional modeling; translated stakeholder requirements into maintainable KPI models and dashboards.
- Built AWS analytics workflows using Athena, S3, Glue, and RDS and implemented validation and data-quality checks to improve downstream reporting reliability.
- Optimized analytical SQL using CTEs, window functions, indexing, aggregation, and caching; automated recurring transformation and reporting workflows with Power Query and DAX.

Scale AI / Outlier AI | AI Data Specialist - Flexible / Task-Based Contract

Apr 2024 - Feb 2025

Remote

- Validated and refined AI-generated datasets and responses for factual accuracy, semantic quality, instruction adherence, and alignment with task requirements and business logic.
- Performed reviewer-level quality checks on peer submissions and provided targeted feedback to improve consistency and output quality.

PROFESSIONAL EXPERIENCE - CONTINUED**JPMorgan Chase | Java Software Engineer - Contract****May 2024 - Oct 2024***Columbus, OH*

- Modernized enterprise banking services using Java and Spring Boot, improving API and database efficiency, maintainability, and deployment reliability in a compliance-sensitive environment.
- Developed secure microservices and integrations using Spring Security, Spring Data JPA, Oracle, Redis, AWS, REST APIs, SFTP, and SMTP; troubleshoot issues across application, integration, and data layers.
- Improved reliability through SQL/query tuning, caching, automated testing, exception handling, Docker/Kubernetes practices, code reviews, CI/CD, and production support.

Investors Bank | Java Software Engineer**Apr 2021 - Dec 2023***Short Hills, NJ*

- Modernized legacy core-banking workflows into Spring Boot microservices and Kafka-based event-driven services using PostgreSQL and Redis while preserving transaction and data integrity.
- Improved backend and database performance through SQL refactoring, indexing, caching, asynchronous processing, and distributed-system troubleshooting.
- Automated build, test, and delivery workflows with Jenkins and GitHub Actions and collaborated with architects, QA engineers, and business stakeholders on incremental modernization.

Erie Insurance | Java Software Engineer**Mar 2019 - Feb 2021***Erie, PA*

- Developed Java backend services and REST APIs supporting claims, policies, customer data, and transaction-processing workflows.
- Implemented concurrency-safe processing, Hibernate/JPA persistence, MySQL query optimization, and automated testing for business-critical services.
- Collaborated with engineering and QA teams on defect resolution, production troubleshooting, application reliability, maintainability, and performance.

SELECTED ENGINEERING PROJECTS**Northbank Banking Platform - Personal Flagship | Distributed Systems & Financial Engineering**<https://github.com/Kalab21/banking-platform>

- Designed and built a Java 17/Spring Boot retail-banking platform with 11 business services plus API gateway and Eureka, Apache Kafka, per-service PostgreSQL databases, Redis, Next.js, Docker, and CI/CD.
- Demonstrates transaction correctness, pessimistic row locking, idempotency, ownership-based authorization, transactional outbox/event-driven workflows, loans/cards, observability, and 1,482 automated tests in CI plus on-demand full-stack suites.

Meridian Lending - Team / FDE | Consumer Lending & Applied AI<https://github.com/2463-FDE/KK-meridian-lending>

- Contributed to brownfield modernization of a multi-service consumer-lending platform covering intake, identity verification, deterministic decisioning, manual review, disclosures, servicing, payments, reconciliation, observability, governance, and a grounded staff-only RAG policy assistant using Python and FastAPI.

Rev-Eval - Team / FDE | Full-Stack Assessment Platform<https://github.com/RevatureFDEPEP/rev-eval/tree/kalabek>

- Extended a FastAPI/Next.js assessment platform with scoring/idempotency, trainer and participant workflows, reporting/analytics, JWT/RBAC, PostgreSQL, MongoDB, MinIO, Nginx, automated testing, and CI/CD.

MarketHub - Personal Project | Java Full-Stack Engineering<https://github.com/Kalab21/markethub>

- Built a Spring Boot MVC marketplace with Admin, Seller, and Buyer workflows using Spring Security, Spring Data JPA, MySQL, Thymeleaf, Docker, GitHub Actions, and 39 automated tests.

EDUCATION & CERTIFICATION

B.S. Computer Science - Maharishi International University - 2023

B.S. Electrical & Computer Engineering - Haramaya University - 2015

Data Analytics Certificate - Per Scholas